ARCHITECTURE READOUT

Agentforce session traces to Databricks Zerobus

How Salesforce Agentforce session traces reach Unity Catalog Delta tables: session export GET, Protobuf POST to Zerobus Ingest OTLP, and where AWS Private Link attaches for Suncorp.

Field	Value
Customer	Suncorp
Scope	Land Agentforce session traces in Unity Catalog Delta via Zerobus Ingest OTLP
Lab environment	Salesforce OrgFarm orgfarm918; Databricks FEVM fevm-ot-pdm-semi (us-east-1); AWS account 332745928618, EKS agentforce-otel (ap-southeast-2)
Prepared by	Pravin Varma
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1. Purpose

This readout describes how Suncorp lands Salesforce Agentforce session traces in Databricks. It sets out the Salesforce session export API, the Zerobus Ingest OTLP contract, the telemetry bridge that connects them, and the point where AWS Private Link attaches. The test plan for the Private Link phase sits in the companion document suncorp-test-setup.md.

1.1 Key definitions

Session export API is Salesforce's Export Agentforce Session Tracing Data endpoint. A client calls GET with one session id and receives OTLP-shaped JSON for that session.

Zerobus Ingest OTLP is the Databricks managed gateway that accepts OTLP Protobuf over HTTPS and writes Unity Catalog Delta tables.

Telemetry bridge is the poller plus a Zerobus client. The poller is the Salesforce client and runs on EKS as CronJob session-poller next to the Collector. The Collector POSTs Protobuf to Zerobus. Phase A1 posted from the laptop straight to Zerobus. The laptop port-forward remains the fallback until an External Client App secret is on the cluster.

2. Background

Suncorp runs customer-facing work on Salesforce Agentforce: LLM reasoning, tool calls, and CRM actions. Platform and risk teams query those traces in Databricks, run evaluations, and retain them with the rest of the lakehouse. Suncorp workspaces use AWS front-end Private Link. The ingest path is Zerobus Ingest OTLP into Delta. Delay follows Data 360 catch-up and the poll interval, typically tens of seconds to a few minutes after the session is readable in Salesforce.

Both products publish the APIs this design uses. This lab wires them together on Databricks and AWS and identifies the network work Suncorp owns for the Private Link phase.

3. Salesforce session tracing

With Agentforce Session Tracing, Einstein Audit, and Agent Platform Tracing enabled, Data 360 stores each session: prompts, tool calls, evaluation spans, metrics, and logs. Salesforce's Export Agentforce Session Tracing Data API returns that payload over REST.

```
GET /services/data/v66.0/einstein/audit/otel/{session-id}
```

A client authenticates with OAuth 2.0 through an External Client App, supplies one session id, and receives JSON that follows the OTLP schema (ResourceSpans, plus metrics and logs). Sessions must have started within the prior 72 hours. Throughput follows Connect API rate limits. Salesforce's developer guidance is to POST this JSON

to the observability backend you operate.

Data 360 DMOs such as ssot__TelemetryTraceSpan__dlm offer a second read path through SOQL. OTLP is the OpenTelemetry Protocol for traces, metrics, and logs over HTTP or gRPC. Zerobus Ingest OTLP accepts that protocol as Protobuf on POST. The Salesforce endpoint returns OTLP-shaped JSON on GET.

4. Databricks Zerobus Ingest

Zerobus Ingest OTLP is a managed gateway. A client POSTs OTLP to {workspaceId}.zerobus.{region}.cloud.databricks.com with Unity Catalog authorization and the header x-databricks-zerobus-table-name. Zerobus writes Delta tables and stores attributes as VARIANT.

The gateway takes Protobuf (application/x-protobuf). The client initiates every POST. The OTLP interface is a documented front-end Private Link service on TCP 443; MQTT and Kafka-compatible Zerobus APIs follow their own networking model. Databricks OAuth tokens last about 60 minutes, so the client refreshes them.

5. The telemetry bridge

Salesforce holds the session and serves it on GET. Zerobus accepts a Protobuf POST and writes Delta. The telemetry bridge is the client of both APIs. Table 1 sets the two contracts side by side.

	Salesforce session export API	Databricks Zerobus OTLP
Job	Session system of record	Lakehouse ingest
Call	Integration GETs OTLP JSON	Integration POSTs OTLP Protobuf
Unit	One session id, 72-hour window	Batches of spans, logs, and metrics
Auth	Salesforce OAuth 2.0 External Client App	Databricks service principal OAuth, Unity Catalog authorization

Table 1. The two API contracts the bridge connects.

The bridge discovers finished sessions through ssot__AiAgentSession__dlm, reads each session with the export GET, converts OTLP JSON to OTLP Protobuf, and POSTs to Zerobus with a refreshed service principal token. Where Salesforce sends an empty spanId on evaluation spans, the converter writes a deterministic 8-byte id so the batch is valid OTLP.

The poller is the Salesforce client. On EKS it is CronJob session-poller in namespace agentforce-otel, POSTing to the Collector ClusterIP. It can also POST Protobuf straight to Zerobus (Phase A1). The Collector is the standing Databricks client for token refresh, batching, and Private Link egress. The CronJob needs SALESFORCE_CLIENT_ID and SALESFORCE_CLIENT_SECRET (External Client App).

6. Architecture

Figure 1 reads left to right. Agentforce writes sessions into Data 360. The poller GETs the session JSON (PULL GET OTel JSON). The poller POSTs Protobuf to the Collector (PUSH Protobuf), and the Collector POSTs to Zerobus (PUSH OTLP Protobuf to Zerobus). Zerobus returns HTTP 200 once the write is durable. Phase A1 used the same Protobuf POST from the laptop with the Collector omitted.

Today the Collector POSTs over public HTTPS to FEVM Zerobus in us-east-1. Phase B keeps the same Collector and Protobuf and points the hostname at a VPC endpoint: front-end or service-direct Private Link on TCP 443 with private DNS. The Salesforce GET uses Salesforce's network path. If that hop must also stay private, Suncorp adds Salesforce Private Connect or an approved egress allowlist as a separate control.

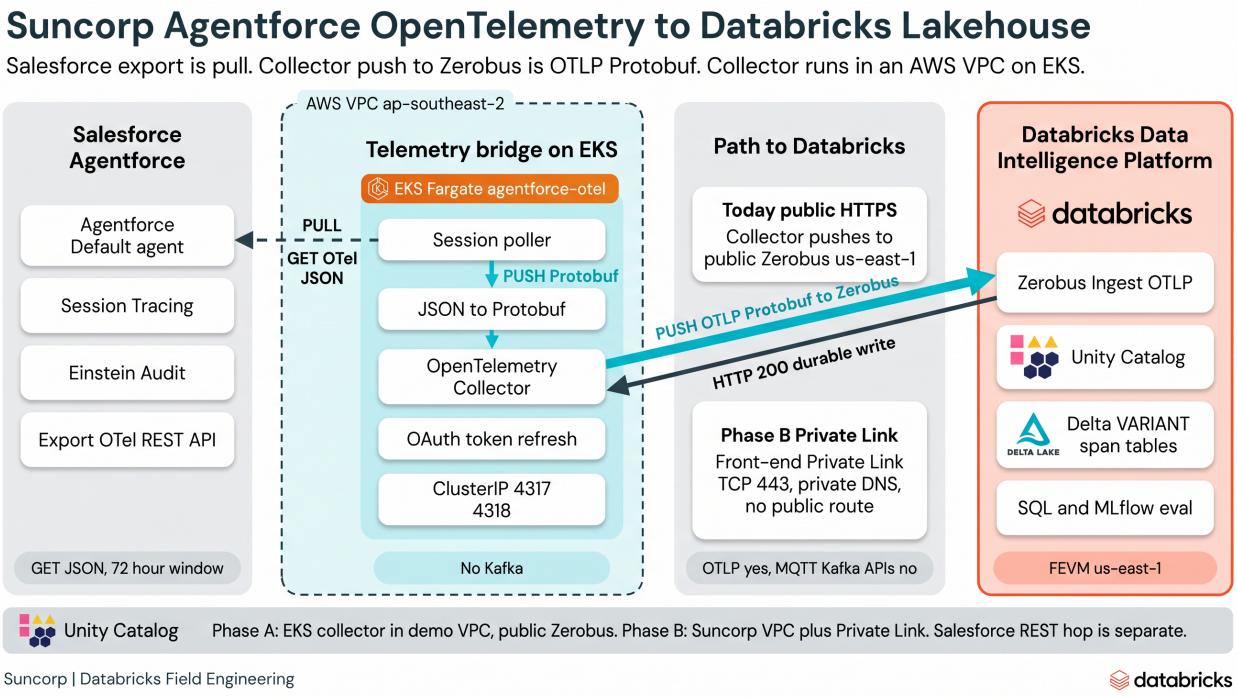


Figure 1. Agentforce session traces to Databricks: session export GET, telemetry bridge on EKS in an AWS VPC, Zerobus OTLP push, and the Phase B Private Link hop.

Component	Where it runs	Role
Agentforce + Data 360 Session Tracing	Salesforce	Runs the agent and stores prompts, tool calls, evaluation spans, metrics, and logs per session.
Session export API	Salesforce	GET /services/data/v66.0/einstein/audit/otel/{session-id} returns OTLP JSON for one session.
Session poller	EKS CronJob session-poller (same namespace as the Collector)	Discovers finished sessions, GETs JSON, converts to Protobuf, fills empty spanIds, POSTs to the Collector ClusterIP. Laptop port-forward is the fallback until an External Client App secret exists.
OpenTelemetry Collector	EKS Fargate agentforce-otel	Receives Protobuf on 4317/4318, batches, sets source attributes, refreshes OAuth, POSTs to Zerobus.
Zerobus Ingest OTLP	Databricks workspace region	Accepts Protobuf with x-databricks-zerobus-table-name and writes Delta tables.
Unity Catalog Delta tables	Databricks	agentforce_otel_spans, _logs, _metrics with VARIANT attributes. SQL and MLflow evals read these tables.

Table 2. Components on the data path.

7. What the lab has demonstrated

Phase	Work	Status
A1	Live Agentforce session 01a16cb0-5079-71ce-a37e-9aee0cb3c332: 8 spans, public Zerobus HTTP 200, rows in Delta.	Done, 24 October 2026
A2	OpenTelemetry Collector in an AWS VPC (lab host: EKS Fargate), POSTing to public Zerobus.	Done, 24 October 2026
B	Same Collector with Private Link: private DNS, VPC endpoint, POST and SQL from the VPC.	Suncorp network build

Table 3. Lab phases and status.

A1 posted Protobuf from the laptop to public Zerobus. A2 runs the Collector in a VPC on the same public hostname. FEVM fevm-ot-pdm-semi is a public workspace. Suncorp copies the Collector-in-VPC pattern onto Private Link. Manifests are in otel-collector/k8s/.

8. Lab and Suncorp production

	This lab	Suncorp production (Phase B)
Salesforce	OrgFarm DE orgfarm918	Suncorp Agentforce + Data Cloud
Databricks	Public FEVM fevm-ot-pdm-semi, us-east-1	Private Link workspace in the same region as the VPC
VPC	Demo VPC with EKS agentforce-otel, ap-southeast-2	Suncorp VPC; Collector uses the Private Link hostname
Collector	Running on Fargate	Same pattern with private DNS and Private Link
Poller	EKS CronJob session-poller; laptop port-forward until ECA exists	Same namespace as the Collector
Catalog	ot_pdm_semi_catalog.agentforce_otel	Suncorp Unity Catalog names, same DDL
Auth	SP agentforce-zerobus-otel	Suncorp SP with table-level SELECT and MODIFY

Table 4. Lab environment against the Suncorp target.

Phase B checks cover private DNS resolution, VPC endpoint connectivity, an OTLP POST returning HTTP 200, and SQL rows for the test session. The detailed cases sit in suncorp-test-setup.md.

9. Produce traces and inspect tables

Catalog ot_pdm_semi_catalog.agentforce_otel. Use the Suncorp Pilot SQL warehouse. DDL is sql/01_create_otel_tables.sql. Repeatable queries are sql/02_demo_queries.sql.

9.1 Produce an Agentforce session

Produce a session: enable Session Tracing and Einstein Audit (dataspace default), then chat with Agentforce (Default) or run sf agent test run --api-name OTel_Demo_Test --target-org orgfarm918 --wait 10 --verbose. Lab utterance: Hello, what can you do? Expected topic: GeneralCRM. Copy generatedData.sessionId.

Wait for Data 360 catch-up (often several minutes). Run DRY_RUN=true python3 poller/poll_agentforce.py. Observe a session id, span count greater than zero, and complete status. Empty spans means Data 360 is still catching up.

9.2 POST to Zerobus

POST Protobuf. Phase A1: set OTEL_COLLECTOR_TRACES_URL to the public Zerobus /v1/traces URL and run the poller with SALESFORCE_SESSION_ID. Phase A2 on EKS: kubectl -n agentforce-otel create job session-poller-once --from=cronjob/session-poller. The Job POSTs to http://otel-collector.agentforce-otel.svc.cluster.local:4318/v1/traces. Observe Exported N span(s) and HTTP 200 in the Job logs. Laptop fallback: kubectl port-forward and POST to 127.0.0.1:4318. Synthetic check: python3 scripts/send_direct_zerobus.py.

9.3 Tables and what to look for

Table	When it grows	What to look for
agentforce_otel_spans	Every successful trace POST	New time; names such as GeneralCRM_*, AiCopilot__ReactInitialPrompt, CLOSED_USER_REQUEST, TaskResolution, InstructionAdherence; populated trace_id and span_id; status.code; attributes VARIANT with prompt and tool payload.
agentforce_otel_logs	When the session export includes log records	Rows sharing the same trace_id; body and severity_text.
agentforce_otel_metrics	When the session export includes metrics	Rows in the same window; metric name and points in VARIANT.

Table 5. Unity Catalog tables after a successful POST.

A short Agentforce chat often fills spans first. Empty logs or metrics with populated spans is a valid session. Lab session 01a16cb0-5079-71ce-a37e-9aee0cb3c332 produced 8 spans and Zerobus HTTP 200. Match a new run by time descending and by name.

```
SELECT time, service_name, name, status.code, trace_id, span_id
FROM ot_pdm_semi_catalog.agentforce_otel.agentforce_otel_spans
ORDER BY time DESC
LIMIT 20;

SELECT name, COUNT(*) AS spans
FROM ot_pdm_semi_catalog.agentforce_otel.agentforce_otel_spans
WHERE time > current_timestamp() - INTERVAL 1 DAYS
GROUP BY name
ORDER BY spans DESC;
```

10. Delivery implication

Suncorp places a Zerobus client (Collector or equivalent) in its VPC, swaps workspace id, catalog, and schema, and keeps Protobuf with x-databricks-zerobus-table-name. Suncorp networking provisions the VPC endpoint and private DNS for Zerobus OTLP. The poller runs in the same VPC and authenticates to Salesforce through an External Client App. VARIANT columns keep prompt and tool payloads available for MLflow evaluation. Delta Sharing can return curated aggregates to Data Cloud once traces land in Delta.